

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2024-2025
FULL MARKS: 75

CSE 4643: Mobile Application Development

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Identify the anti-patterns present in the following scenario and propose suitable design improvements for each issue.

10
(CO3)
(PO2)

A well-known software development company, recently launched a mobile application called Medication Tracker to help users manage their daily medications, receive reminders, monitor dosage history, and communicate with healthcare providers. To simplify navigation, the development team implemented a side drawer navigation menu that is accessible from a button located at the bottom of the screen. The drawer contains more than fifteen primary categories, including Medication Schedule, Reminders, Prescription History, Doctor Contacts, Health Reports, Settings, Emergency Support, and Community Forums. The categories are arranged in no particular order, with some frequently used features mixed with less important ones. Although the most critical features are placed near the top of the drawer, users often need to scroll through a long list to locate specific options. In addition, several menu items use technical medical terminology that many users do not understand. Some categories contain submenus that are hidden until tapped, while others open new screens immediately, leading to inconsistent behavior. User feedback indicates that elderly users frequently struggle to find important functions such as medication reminders and emergency contacts, especially when using the application in stressful situations.

- b) Find the output of the following Code Snippet 1 with brief justification.

5 + 5
(CO2)
(PO1)

```
1 const arr = [1, '1.5', 1.6];
2
3 const arr_manipulation = (ele) => {
4   if (typeof ele === 'string') {
5     console.log(parseFloat(ele) * 2 + 2);
6   }
7   else if (typeof ele === 'number' && Number.isInteger(ele)) {
8     console.log(ele * 1.5 + 10);
9   }
10  do {
11    console.log('Looping '+ele+' Times.')
12  } while (ele < 0);
13
14  }
15  else if (typeof ele === 'number') {
16    console.log(parseInt(ele) * 3);
17  }
18
19  console.log('Summation of '+ele+' with 9 is: '+(ele+9));
20 };
21 arr.forEach(arr_manipulation);
```

Code Snippet 1: A Java Script program for Question 1.b

- c) Define and illustrate the Skeuomorphic Mobile Design Pattern with a clearly labeled diagram.
2. a) Complete the Styling block (Lines 20-22) in Code Snippet 2 so that the output matches the design shown in Figure 1.

```

1 import { View, StyleSheet } from 'react-native';
2
3 export default function App() {
4   return (
5     <View style={styles.container}>
6       <View style={styles.boxA}>A</View>
7       <View style={styles.boxB}>
8         <View>B.1</View>
9         <View>B.2</View>
10      </View>
11      <View style={styles.boxC}>
12        <View>C.1</View>
13        <View>C.2</View>
14      </View>
15      <View style={styles.boxD}>D</View>
16    </View>
17  );
18 }
19
20 const styles = StyleSheet.create({
21   // Complete this block
22 });

```

Code Snippet 2: A React Native program for Question 2.a

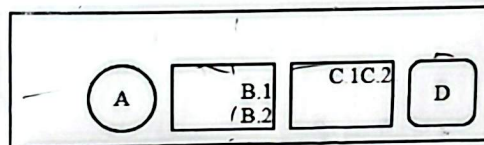


Figure 1: A figure for Question 2.a

- b) Illustrate the most suitable navigation pattern for the following application scenario with a clearly labeled diagram and provide a clear justification for your selection.
- IUTPS plans to launch a mobile application for showcasing photographs. The application will consist of three primary categories: SHOOT-AT-IUT, Contests, and Winners. These categories are of equal importance and should remain readily accessible to users at all times. Additionally, the title SHOOT-AT-IUT may change dynamically (e.g., SHOOT-ON-16December) to reflect the current contest being organized. The application content will be updated frequently. As the primary purpose of the application is photo presentation, the user interface should remain simple, intuitive, and efficient.*
3. a) Discuss the following three concepts in the context of React Native, providing suitable examples for each.
- Asynchronous Programming
 - Custom Components
 - Conditional Rendering of React Native

- b) Identify and correct the errors present in Code Snippet 3 with brief justification within the context of the React Native framework.

```
1 import React, { useState, useEffect } from 'react';
2 import { View, Text, TextInput, ScrollView, Button } from 'react-
  native';
3
4 export default function App() {
5
6   const [name, setName] = useState('');
7   const [count, setCount] = useState(0);
8
9   useEffect(() => {
10     setCount(1);
11   }, [count]);
12
13   const increment = () => {count = count + 1;}
14   const fruits = ['Apple', 'Banana', 'Mango'];
15
16   return (
17     <View style={{ flex: 1 }}>
18       <TextInput value={name} onChangeText={(text) => setName(text)}
19         />
20
21       <Button title="Increment" onPress={increment()} />
22
23       <ScrollView horizontal={true}>
24         {fruits.map((fruit, index) => {
25           <Text key={index}>{fruit}</Text>
26         })}
27
28       </ScrollView>
29
30       <Text>Total Count: count</Text>
31     </View>
32   );
33 }
```

Code Snippet 3: A React Native program for Question 3.b